OSI Model Overview

The OSI model is a conceptual framework that standardizes communication into seven layers.

7	Application Layer
6	Presentation Layer
5	Session Layer
4	Transport Layer
3	Network Layer
2	Data Link Layer
1	Physical Layer



Close to user

Deals with user applications and data representation.

Close to hardware

Deals with physical transmission of bits.

Memory Trick:

All People Seem To Need Data Processing

Key Takeaway:

The seven layers help organize communication tasks from the user down to the hardware.

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OSI Layers Explained

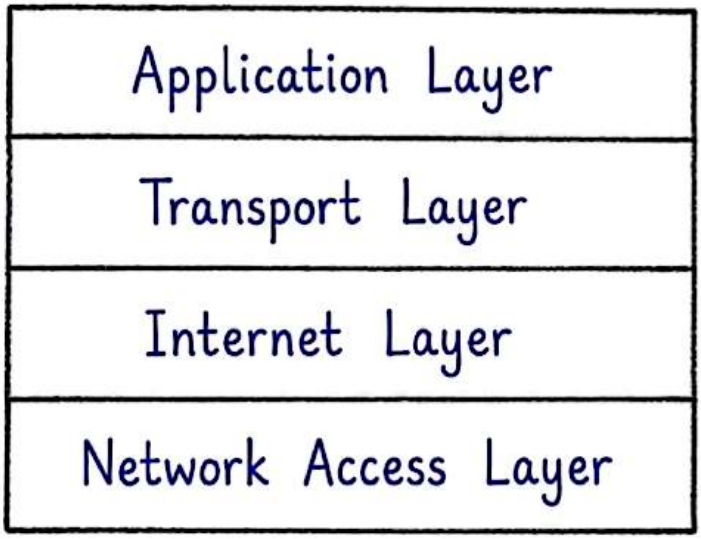
Layer	Main Function	Example
Application	Provides services to user	HTTP, FTP, SMTP, DNS
Presentation	Data format, encryption, compression	JPEG, ASCII, SSL
Session	Establish, manage, close sessions	NetBIOS, RPC
Transport	End-to-end delivery, reliability	TCP, UDP
Network	Logical addressing and routing	IP, ICMP
Data Link	Framing, MAC addressing, error control	Ethernet, PPP
Physical	Transmit raw bits over physical medium	Cables, Signals

Key Point: Each layer uses services from the layer below and provides services to the layer above.

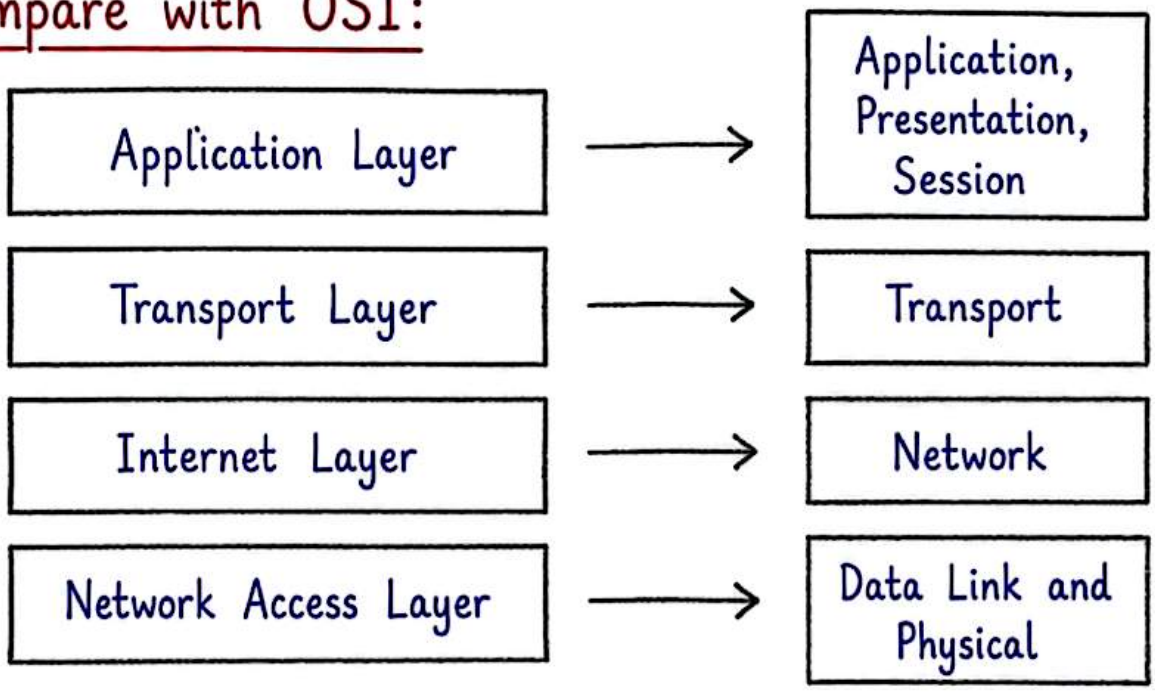
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TCP/IP Model

TCP/IP is the practical model used on the Internet.



Compare with OSI:



TCP/IP is simpler and suited for real-world implementation.

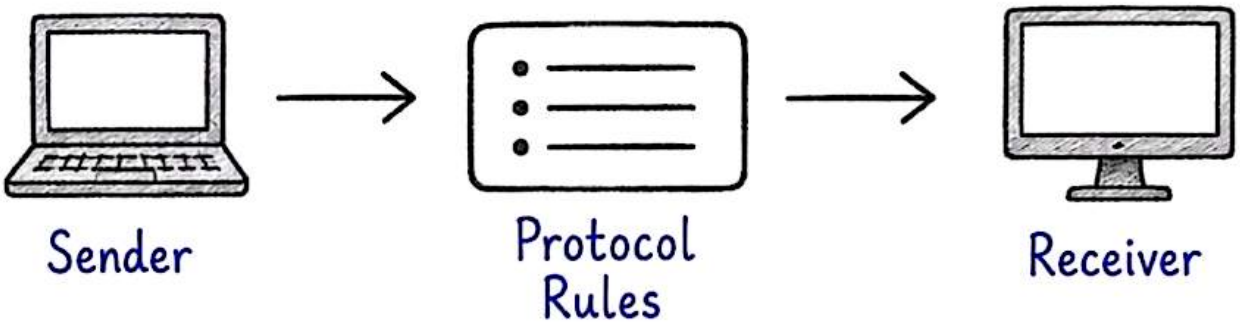
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Protocols

A protocol is a set of rules and standards that governs how data is transmitted and received between devices.

Examples:

- HTTP – Web browsing
- HTTPS – Secure web browsing
- FTP – File transfer
- SMTP – Email sending
- DNS – Domain name resolution
- TCP – Reliable data transfer
- UDP – Fast data transfer
- IP – Addressing and routing



Protocols make communication organized and understandable.

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TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable	Faster
Ordered delivery	No reliability guarantee
Flow control	No order guarantee
Error checking and recovery	No flow control
Header size 20-60 bytes	Error checking but no recovery
Used in web, email, file transfer	Header size 8 bytes
Used in web, email, file transfer	Used in streaming, gaming, video calls

TCP is reliable but slower.
UDP is fast but less reliable.

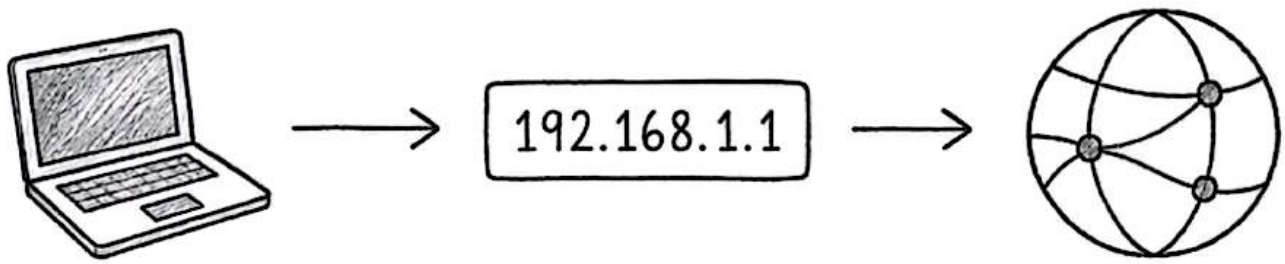
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IP Address

An IP address uniquely identifies a device on a network.

Types:

- IPv4 – 32-bit address, example 192.168.1.1
- IPv6 – 128-bit address
- Public IP – used on the Internet
- Private IP – used in local networks
- Static IP – manually assigned
- Dynamic IP – assigned by DHCP



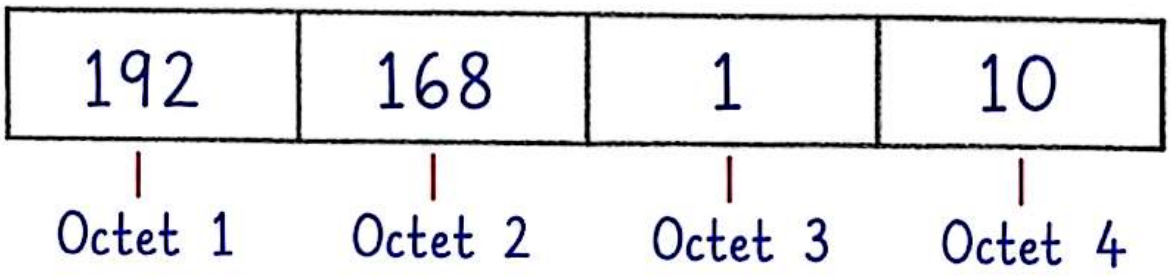
Final Takeaway:
IP address gives a device a unique network identity.

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IPv4 Addressing

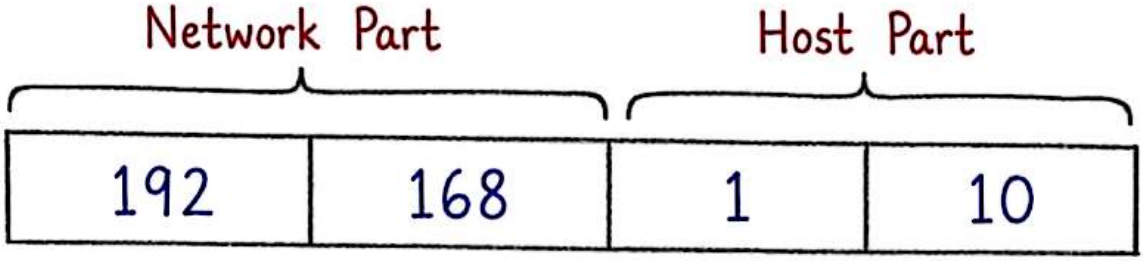
IPv4 is a 32-bit address written in dotted decimal notation.

192.168.1.10



Parts:

- Network Part – identifies the network
- Host Part – identifies the device



Note:

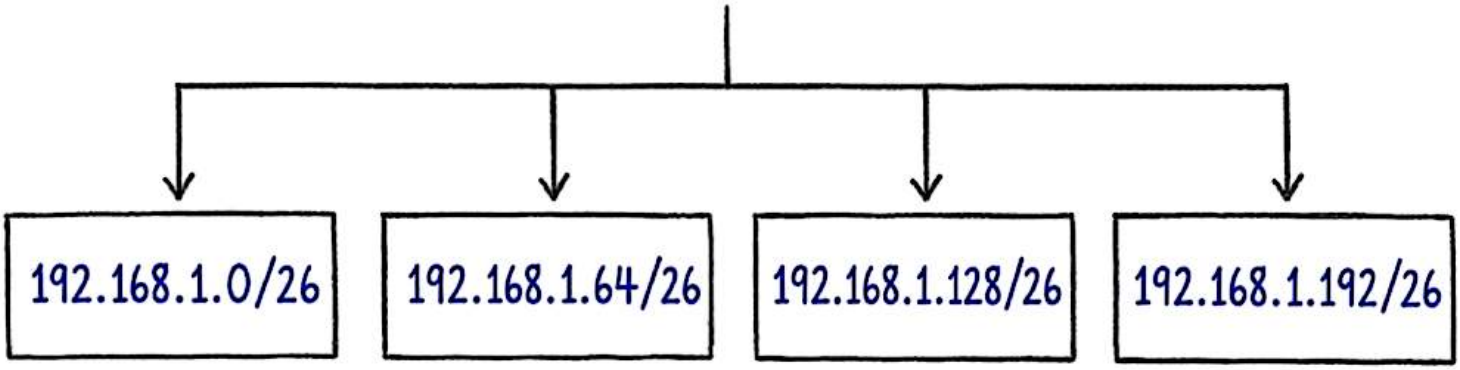
- 1 octet = 8 bits
- 32 bits = 4 octets

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Subnetting Basics

Subnetting divides a large network into smaller networks for better management and security.

Example Network: 192.168.1.0/24



Key terms:

- Subnet Mask – defines network and host parts
- Network ID – first address of network
- Host ID – identifies devices
- CIDR Notation – /24, /26, /30 etc.

Final Takeaway:
Subnetting improves organization and security.

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MAC Address

A MAC address is a unique physical address assigned to a network interface card.

Features:

- 48-bit address
- Written in hexadecimal
- Used at Data Link Layer
- Used within local networks
- Example: A4:5E:60:12:AB:90

Difference:

IP Address – Logical Address

MAC Address – Physical Address

MAC address identifies hardware inside a local network.

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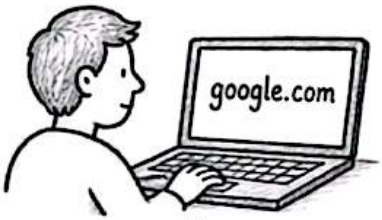
DNS

DNS converts human-readable domain names into IP addresses.

google.com ⟶ 142.250.190.14

How DNS works:

① User types domain name



② DNS Resolver



The resolver asks a DNS server for the IP address.

③ DNS Server



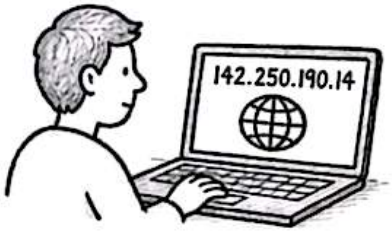
The DNS server looks up the domain and finds the IP address.

④ Returns IP Address



The IP address is sent back to the resolver.

⑤ Website loads



The browser uses the IP address to connect to the website and load the page.

Final Takeaway:

DNS is like the phonebook of the Internet.

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